Technical Report

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Device and network design

Part 1: CANopen physical layer

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This version replaces version 2.0.0. The main changes are as follows:

- minor editorial improvements.

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Introduction

In order to establish a reliable CAN interconnection, it is necessary to consider different device and network design aspects. The recommendations given in this document are dedicated for use in CANopen networks. This includes some general design guidelines on wiring harness (cables, connectors, topologies, and location of termination resistors).

Through the years, CiA has recommended the pin-assignment for many kinds of connectors, which have been published in previous versions of this document. In order to generalize these recommendations, CiA has moved the pin-assignment recommendations to the CiA 106 document, which is not more related to classic CANopen only.

Contents

1 Scope

This document provides device and network design recommendations for the CANopen physical layer. Additionally, it provides guidelines for selecting cables for use in CANopen systems.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

AN 96116:1996, *Application note PCA82C250/251 CAN Transceiver, NXP*

CiA 106, *Connector pin assignment recommendations*

ISO 11898-2, *Road vehicles - Controller area network (CAN) - Part 2: High-speed medium access unit*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in the CiA terminology database (https://www.can-cia.org/groups/cia-glossary-of-terms/) and the following apply.

3.1
bus cable
cable terminated at both ends by termination resistors

3.2
stub cable
short branch of a bus cable not terminated by a resistor and connected to the CANopen device

3.3
T-connector
T-shape electric connector with three connection points for a bus cable

3.5
trunk cable
backbone of the bus cable without stub cables

Additionally, ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp/

— IEC Electropedia: available at http://www.electropedia.org/

4 Symbols and abbreviated terms

For the purposes of this document, the following symbols and abbreviated terms apply.

AC	alternating current
DC	direct current
SJW	resynchronization jump width

5 AC and DC parameters

5.1 Bus cable and termination resistors

The bus cables, connectors, and termination resistors should match regarding the impedance. Impedance mismatches cause reflections, which can lead to communication errors. The stub cable should be as short as possible. The socket connector may be powered. The plug connector should not be powered; this is the reason why most devices are equipped with plug connectors. The CANopen devices can be connected to the network either directly to the T-connector or with a stub cable. T-connectors provide an easy removal of a CANopen device without disrupting network operation.

Table 1 shows some recommended values for CANopen networks with less than 64 nodes (source: Table 4 and Table 5 in AN 96116:1996).

Table 1 – Recommended values for CANopen networks

Bus length [m]	**Bus cable**[a]		**Termination resistance [Ω]**	**Bit rate [kbit/s]**
	Length-related resistance [mΩ/m]	**Cross-section**[b] **[mm²]**		
0 to 40	70	0,25 to 0,34	124	1000 at 40 m
40 to 300	<60	0,34 to 0,6	150 to 300	≤500 at 100 m
300 to 600	<40	0,5 to 0,6	150 to 300	<100 at 500 m
600 to 1000	<26	0,75 to 0,8	150 to 300	<50 at 1 km

[a] Recommended cable parameters: 120-Ω impedance and 5-ns/m specific line delay.

[b] The wire cross-section for stub cables should be between 0,25 mm^2 and 0,34 mm^2.

Besides the cable impedance, the actual impedance of the connectors should be considered, if calculating the voltage drop. The transmission resistance of one connector should be in the range of 2,5 mΩ to 10 mΩ.

AN 96116:1996 recommends the following values:

- minimum dominant value $V_{\text{diff.out.min}}$ = 1,5 V,
- minimum differential resistance $R_{\text{diff.min}}$ = 20 kΩ,
- requested differential input voltage $V_{\text{th.max}}$ = 1,0 V,
- minimum termination resistance $R_{\text{T.min}}$ = 118 Ω.

Table 2 recommends the maximum wiring length given for different bus cables and different number (n) of connected bus nodes (source: Table 6 in AN 96116:1996).

Table 2 – Maximum wiring length

Wire cross-section [mm²]	**Maximum length [m]** [a]			**Maximum length [m]** [b]		
	***n* = 32**	***n* = 64**	***n* = 100**	***n* = 32**	***n* = 64**	***n* = 100**
0,25	200	170	150	230	200	170
0,5	360	310	270	420	360	320
0,75	550	470	410	640	550	480

[a] safety margin of 0,2 m

[b] safety margin of 0,1 m

If driving more than 64 nodes using bus length of more than 250 m, the V_{CC} supply voltage for the ISO 11898-2 transceiver should have an accuracy of 5 % or better. The minimum supply voltage should be at least 4,75 V when driving a 50-Ω load, i.e. 64 nodes, and at least 4,9 V when driving a 45-Ω load i.e. 100 nodes.

5.2 Stub cable

As a rule of thumb, the following relation should be used for a single stub cable length l_{u} (source: AN 96116:1996):

$$l_u < \frac{t_{PROPSEG}}{50 \cdot t_p} \qquad [1]$$

with the specific line delay per length unit t_{p} = 5 ns/m and the time of the propagation segment

$$t_{PROPSEG} = t_{SEG1} - t_{SJW} \qquad [2]$$

Additionally, the cumulative stub length l_{ui} should be calculated with the following formula:

$$\sum_{i=1}^{n} l_{ui} < \frac{t_{PROPSEG}}{10 \cdot t_p} \quad [3]$$

This leads to a reduction of the maximum stub cable length by the sum of the actual cumulative stub cable length at a given bit rate. If the above recommendations are met, then the probability of reflection problems is considered to be low.

5.3 CAN ground and galvanic isolation

In complete galvanically-isolated CANopen networks, the CAN ground signal should be carried in the cable line. It should be connected at only one point to the CAN ground potential. If one CANopen device with a not galvanically-isolated interface is connected to the network, the connection with the CAN ground potential is given. Therefore, only one device with a not galvanically-isolated interface should be connected to the network.

The network designer is responsible to guarantee that the common mode rejection of the transceivers has still reached the upper limit.

5.4 External power supply

The output voltage at the optional power supply should be +18 V_{DC} < V_{CC}+ < +30 V_{DC} in order to enable the use of +24-V_{DC} power supplies.

